DISSERTATION PROPOSAL PRESENTATION

SmartHelp AI

Intelligent Complaint Classification and Routing System

Python · FastAPI · React · PostgreSQL · scikit-learn



Presented by	[Student Name] · [BITS ID]
Supervisor	[Supervisor Name], [Designation]
Organisation	[Organisation / Mentor Name]

BITS Pilani · CSIW ZG628T Dissertation · Academic Session 2026–2027

OVERVIEW

Presentation Outline

What we will cover in the next 10–15 minutes

01 Problem and motivation

02 Proposed solution

03 Objectives and scope

04 Architecture and workflow

05 AI / NLP methods

06 Database design

07 Evaluation plan

08 Timeline and conclusion

The Real-World Problem: Complaint Triage

A busy campus helpdesk must read and route every request

- Staff read and categorise every complaint by hand
- Wrongly assigned complaints need reassignment
- Repetitive first-step work slows responses
- Hard to trace the original routing decision

PROBLEM STATEMENT

How can NLP suggest complaint categories so requests reach the right department, while staff keep oversight of every decision?

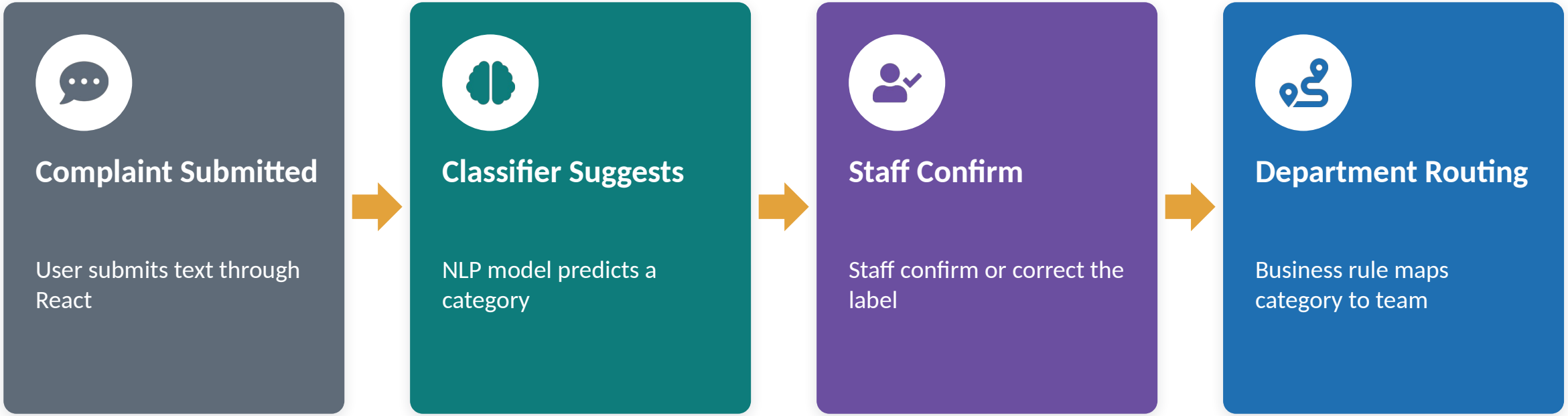
Why it matters

Example: "The fee was deducted but the portal still shows pending" — which department handles this?

OUR PROPOSED SOLUTION

From Manual Routing to AI Assistance

We automate the repetitive first step, not human responsibility



In one sentence: Use NLP to suggest complaint categories, then route requests with staff oversight — AI suggests, rules route, people decide.

OBJECTIVES

What We Propose to Build

Three groups of proposed features

Application

- Registration and login
- Complaint submission and tracking
- Staff status updates
- Basic admin analytics

AI component

- TF-IDF text features
- Compare three classifiers
- AI-suggested category

Oversight

- Staff correction of labels
- Department routing rules
- Prediction + audit history

RESEARCH CORE

an NLP category suggestion with human-in-the-loop correction and full traceability.

SCOPE

What Is In — and Out — of the First Release

Scope control keeps the dissertation achievable

In scope

- User registration and login
- Complaint submission and tracking
- AI-suggested category
- Department routing
- Staff correction and status updates
- Basic admin analytics

Out of scope / future work

- Paid generative-AI APIs or chatbots
- Voice or image complaints
- Multilingual classification
- External helpdesk integrations
- Automatic complaint resolution

The AI suggests a category — it never resolves complaints or replaces staff judgement.

SYSTEM ARCHITECTURE

How the Application Components Communicate

Proposed layered design

User / Staff / Admin — Three roles with different screens

↓

React frontend — Complaint forms, dashboards, status display

↓

FastAPI backend — REST APIs, authentication, validation, routing rules

↓

Python NLP model — TF-IDF + selected classifier, loaded inside the backend

PostgreSQL

Users, complaints, AI predictions, assignments and status history

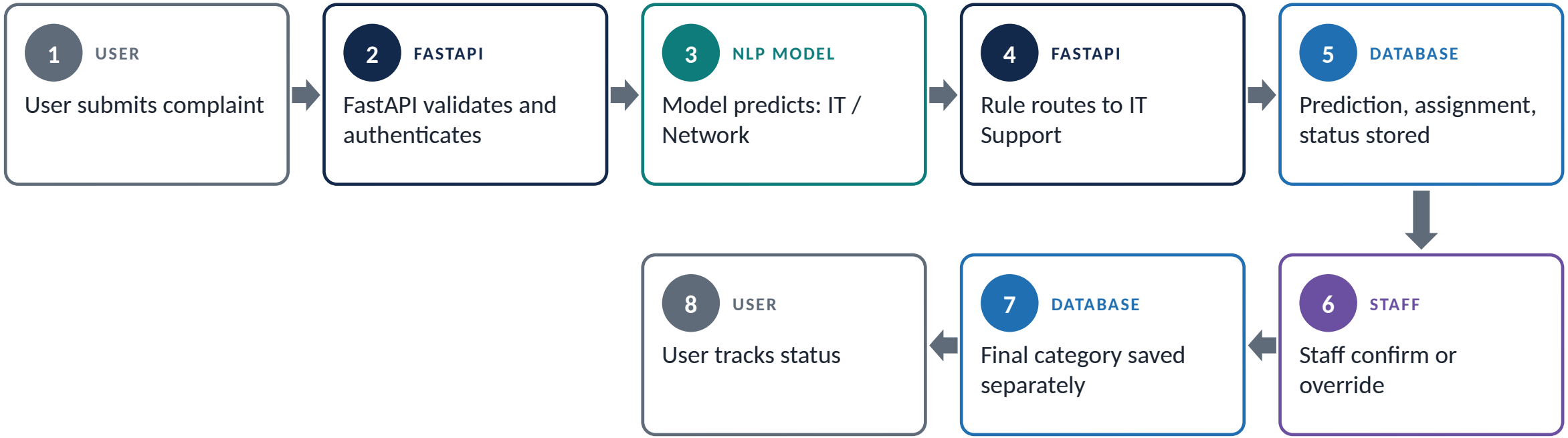
Trust

React never queries the database or sends its own AI labels

END-TO-END WORKFLOW

One Complaint: End-to-End

Illustrative example — "Wi-Fi is down in Lab B"



Design rule: the original AI prediction is never overwritten — staff corrections are stored separately with history.

AI METHOD 1

Text Classification and TF-IDF

Supervised learning from labelled complaints

HOW IT WORKS

- Input X = complaint text; label y = category
- TF-IDF turns words into weighted numbers
- Common words (e.g. "the") get low weight
- TF-IDF is fitted on training data only

Illustrative training rows (not a real dataset)

Complaint text	Category
"Campus internet keeps dropping"	IT / Network
"Amount deducted; receipt unavailable"	Accounts / Billing
"Classroom projector is damaged"	Infrastructure

Real category labels must come from a selected, documented dataset — still a pending decision.

AI METHOD 2

Candidate Classifiers

Three lightweight models compared on real evidence

TRAINING AND SELECTION

- Train on labelled complaints; test on held-out data
- Compare with macro-F1 because categories may be imbalanced
- Save the selected pipeline with a model version
- Load it in FastAPI to predict new complaints

Naive Bayes
Probabilistic baseline — a useful first benchmark

Logistic Regression
Learns weights linking text features to categories

Linear SVM
Learns separating boundaries — strong for sparse text

HUMAN IN THE LOOP

Responsible AI: Staff Can Correct Predictions

An AI prediction is a suggestion, not the final truth

WHY IT MATTERS

- Complaints can be ambiguous (e.g. "login fails on Wi-Fi")
- Staff confirm or correct every uncertain label
- We store both the AI label and the final label
- Corrections show where the model needs improvement

POST /api/v1/complaints · illustrative

```
Request:
{ "description": "Wi-Fi is down in Lab B" }

Response:
{ "id": "C-101", "status": "SUBMITTED" }
```

Stored separately: AI-predicted category + model version, and the staff-approved final category with who changed it and when.

DATABASE DESIGN

What PostgreSQL Will Store

Conceptual entities — final schema designed during implementation

USERS
role · credentials

COMPLAINTS
text · owner · status

AI_PREDICTIONS
label · model version

CATEGORIES
approved labels

DEPARTMENTS
service teams

ASSIGNMENTS
routing · owner

STATUS_HISTORY
who · what · when

MODEL_VERSIONS
trained artifact

KEY RELATIONSHIPS

- A complaint belongs to one user
- Each complaint has an AI prediction with a model version
- Categories map to departments
- Assignments track which team owns a complaint
- Status history records every change
- Predictions are never silently overwritten

EVALUATION PLAN

How We Will Evaluate the System

Planned methodology — no experiments have been run yet

ML EVALUATION

- Accuracy and per-class metrics
- Precision and recall
- Macro-F1
- Confusion matrix
- Error analysis

ALSO TESTED

Role-based access, valid/invalid input, prediction persistence, staff override and routing (Pytest).

Model comparison

RESULTS PENDING

Model	Accuracy	Macro-F1
Naive Bayes	TBD	TBD
Logistic Regression	TBD	TBD
Linear SVM	TBD	TBD

There are no experimental scores yet. Model quality and software correctness will be reported separately — a good F1 score does not prove the app is secure.

PROJECT PLAN

Implementation Timeline and Milestones

Proposed 16-week work plan



2 Dec 2026
Mid-semester report

20 Jan 2027
Final report

Week 16
Final demonstration

CONCLUSION

Expected Outcomes and Limitations

What we expect to deliver — and what we will not claim

Expected outcomes

- A working complaint-triage web prototype
- NLP category suggestions with staff oversight
- A measured comparison of three classifiers
- Full prediction and status history

Limitations

- English text and a fixed label set only
- Dataset source still to be finalised
- Ambiguous complaints need human review
- Not production-ready without a security review

SmartHelp AI automates the repetitive first step of triage while keeping people responsible for every decision.

THANK YOU

Thank You — Questions?

We welcome feedback from our supervisor and reviewers

KEY TAKEAWAYS

1. AI suggests the category
2. Business rules route the request
3. People confirm or correct
4. Every decision is stored and traceable

PRESENTED BY

[Student Name]
[BITS ID]
[Email address]

SUPERVISOR

[Supervisor Name]
[Organisation]

SmartHelp AI is a proposed project. All examples, screens and values in this presentation are illustrative.